

BLOCK 6

RS project

Capstone project studio

Capstone purpose

Design and evaluate a recommender-system solution for a clearly defined problem.

Problem

Dataset

Baseline

Method

Evaluation

Reflection

Group size and weekly milestones

Group size

3–4 students
Target: 8-10 groups

Day	Milestone
Mon	group, topic, dataset, baseline idea
Tue	problem statement and data audit
Wed	method and preliminary experiment plan
Thu	evaluation review and risks
Fri	4-minute project proposal pitch

Suggested project tracks

Core ranking

compare classical methods

Sequential RS

predict next item/session

Multimodal/content

use text, metadata or images

Trustworthy RS

diversity, fairness, bias

Retrieval & scale

candidate generation

LLM / agentic RS

controlled candidate set + baseline

Minimum project requirements

- Target user and application context
- Open dataset and feedback type
- Recommendation task and candidate set
- One simple baseline + one improved method
- Reproducible split and evaluation protocol
- At least two relevance metrics + one beyond-accuracy metric
- One limitation, risk or failure mode

Research question template

Under evaluation protocol P, does method X improve metric Y over baseline Z, and what trade-off does it create for metric W?

Example

Does a sequential model improve NDCG@10 over a popularity baseline on MIND-small, and does it reduce catalogue coverage?

Friday pitch format

Slide 1

Problem and target user

Slide 2

Dataset and feedback signals

Slide 3

Research question, baseline and method

Slide 4

Evaluation protocol, metrics and risks

Timing: 4 minutes presentation + 1 minute question/transition

Project grading criteria

Criterion	Weight
Problem framing and relevance	10%
Data understanding and reproducibility	15%
Meaningful baseline	15%
Method design and implementation	20%
Evaluation rigor	25%
Analysis, limitations and responsible use	10%
Communication quality	5%

Final deliverables and deadline

Submission deadline: 28 September 2026

- Scientific report, max. 10 pages, single-column
- Code submission or repository link; data does not need to be submitted
- Clear contribution statement for each group member
- Honest report on the responsible use of AI (coding) tools
- Short learning reflection, max. 1 page, excluding the 10-page limit

Project studio: first decisions today

Domain

What user problem are you solving?

Dataset

What feedback and item data exist?

Baseline

What is the simplest comparison?

Metric

What does success mean?

Output before leaving: group list + tentative project track + dataset idea